

Preparation and rheological study of pentaerythritol triacrylate grafted onto polypropylene induced by air plasma

Xiao-Mei Meng, Zhe Ma, Gui-Qiu Ma , Jing Sheng

School of Materials Science and Engineering, and Tianjin Key Laboratory of Composite and Functional Materials, Tianjin University, Tianjin 300072, China

Correspondence to: Z. Ma (E-mail: zhe.ma@tju.edu.cn) and G.-Q. Ma (E-mail: magq@tju.edu.cn)

ABSTRACT: The surface modification of isotactic polypropylene by dielectric barrier discharge of air plasma is studied. The air plasma induces melt grafting of pentaerythritol triacrylate onto polypropylene, which successfully introduces polar long-chain branching (LCB) structure into polymer bulk. The results of X-ray photoelectron spectroscopy show that C—O and C=O oxygen-containing functional groups are formed on polypropylene surface, of which the functional groups contents can reach 24.4 and 12.0%, respectively. It is found that there is an optimum plasma treatment time of 4 min to obtain the high peroxides concentration of about $3.38\text{--}3.69 \times 10^{-7} \text{ mol cm}^{-2}$. The highest grafting degree of this plasma method is 1.79%, which achieves the same level as the chemical grafting method. The molecular structural change of the graft-modified polypropylene, in the form of LCB is investigated by shear rheology. © 2019 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2019**, 136, 48054.

KEYWORDS: grafting; irradiation; polyolefins; rheology; synthesis and processing techniques

Received 26 February 2019; accepted 19 May 2019

DOI: 10.1002/app.48054

INTRODUCTION

Isotactic polypropylene (iPP) is one of the most important polyolefin materials due to its excellent physical and chemical properties, such as low density ($0.89\text{--}0.91 \text{ g cm}^{-3}$), high melting point ($164\text{--}170^\circ\text{C}$), good mechanical properties (tensile strength of about $33\text{--}40 \text{ MPa}$), and excellent chemical resistance (insoluble in most organic solvents at room temperature). However, iPP is the typical nonpolar material, it has poor compatibility with other polar materials (e.g., nylon, polyurethane, and polycarbonate), which limits the broad applications of iPP materials in packaging. Therefore, polypropylene modification has been developed for decades.^{1–3} As a green and environment-friendly method, plasma technology has many unique advantages in polymer modification, like operation at room temperature, absence of catalyst and initiator, low pollution. In recent years, the use of various plasma techniques to modify polymer has attracted more and more attention, which include plasma surface treatment as well as plasma-induced graft copolymerization.^{4–7} The energetic particles of plasma interact with polymer, which can result in four major effects on surface, including cleaning, ablation/etching, crosslinking, and chemical modification. If oxygen is involved, the plasma-modified polymer surface can be oxidized to form peroxides, which can further induce graft copolymerization of the monomer, in particular with some functional monomers,^{8–11} like maleic anhydride, acrylic acid, styrene (St), and methyl methacrylate (MMA). Generally, plasma-induced graft copolymerization is carried out in two steps. The matrix

material is first activated by plasma treatment and then contacted with the monomer and chemical initiator to initiate graft copolymerization. The plasma-induced grafting can be implemented with solution grafting, vapor phase grafting, and melt grafting. Compared with the traditional solution grafting and vapor phase grafting, melt grafting has two advantages. One is that the melt grafting requires much less time, around tens of minutes, than hours in solution or vapor phase grafting.^{12,13} The other is that melting grafting achieves the modification of polymer bulk, more efficient than the only surface modification in solution grafting and vapor phase grafting.^{14,15} There are few reports on the melt grafting modification of materials by plasma-induced polymerization. Meemusaw *et al.*¹⁶ studied the air plasma-induced graft copolymerization of MMA onto the high-density polyethylene by reactive extrusion. Actually, when the introduced grafting chain is long enough, melt strength, and processability of grafted materials can be improved. In the meanwhile, the rheological properties of a melt are strongly affected by the presence of long-chain branching (LCB), including viscosity, viscoelasticity elasticity, and activation energy. For this reason, many researchers have studied the microstructural information of the LCB polymers by rheological methods.^{17–19}

In this work, the surface modification of polypropylene was studied using dielectric barrier discharges (DBDs) of air plasma. The introduction of polar and LCB to the polypropylene bulk was successfully realized by air plasma technology for the first time.

The effects of plasma treatment conditions and mixing conditions on grafting degree (GD) were discussed in detail. The incorporated polar groups can facilitate the blending of polypropylene with other polar polymers. In addition, the resultant LCB is beneficial to expand the applications of polypropylene in foaming and blow molding.

EXPERIMENTAL

Materials and Equipment

The *i*PP (T30s) was supplied by Lanzhou Petrochemical Corporation, Lanzhou, China, of which the melt flow index is 2.8 g/10 min (2.16 kg at 230 °C). Pentaerythritol triacrylate (PETA, CP) was obtained from Adamas Co. (Shanghai, China) and the air (AR, 99.999%) was obtained from Tianjin Liufang Gas Co. (Tianjin, China). The 2,2-diphenyl-1-picrylhydrazyl (DPPH, AR) was supplied by TCI Co. (Shanghai, China). All reagents were used without further purification.

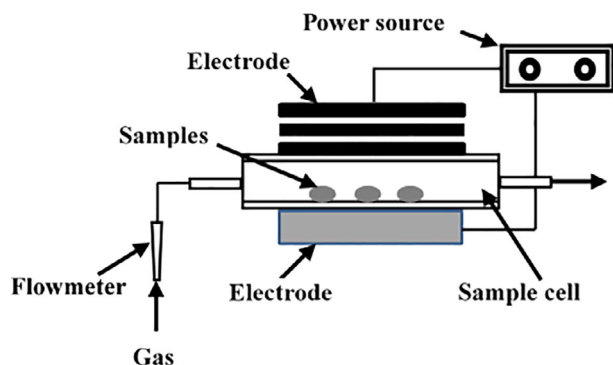
Plasma modification was implemented using a DBDs plasma apparatus shown in Scheme 1. The plasma excitation power supply employed was a high-voltage power supply (CTP-2000K; Nanjing Suman Electron Co., Ltd., Nanjing, China) with a variable discharge voltage (from 0 to 150 V) and an invariable frequency (10 kHz).

Plasma Surface Modification

The *i*PP pellets were compression molded at 180 °C to sheets with the thickness of 1 mm, which were cut into the square samples with the dimension of 10 × 10 mm². The surface of *i*PP was modified by air plasma at room temperature under atmospheric pressure. The airflow rate was fixed at 100 mL min⁻¹. Four discharge voltages of 55, 70, 85, and 100 V were selected. The surface treatment times of air plasma include 1, 2, 4, 8, and 12 min.

Peroxides Determination

The amount of peroxides on the surface of plasma-treated *i*PP (PiPP) was determined by DPPH method.^{20,21} The PiPP samples were immersed in ethanol solution of DPPH (1.0 × 10⁻⁴ mol L⁻¹, 10 mL) and kept at 70 °C for 24 h to decompose the peroxides formed on and near the sample surface. The DPPH consumptions were determined by the difference of absorbance at 520 nm between *i*PP and PiPP samples using a UV-Vis spectrometer (Cary 60 UV-Vis; Agilent, Shanghai, China).



Scheme 1. Illustration of plasma-induced surface modification system. [Color figure can be viewed at wileyonlinelibrary.com]

Graft Copolymerization

The *i*PP pellets were directly treated by air plasma, where discharge voltages were varied (55, 70, 85, and 100 V) and treatment time was fixed at 4 min. The airflow rate was chosen at 100 mL min⁻¹. After plasma treatment, the grafting reaction of PETA onto PiPP was carried out in a torque rheometer (XXS-300 mixer; Shanghai Kechuang Rubber Machinery Equipment Co., Ltd., Shanghai, China). The reaction recipe contains the fixed amount of PiPP (45 g) and variable PETA (0–5 wt %). The mixing rate (rotation speed) was set at 60 rpm and mixing temperatures were 170, 180, 190, and 200 °C, respectively. The reaction time was set for 10, 15, and 20 min, respectively.

Purification of Sample

The reacted samples were dissolved in xylene at 140 °C and then the solutions were poured into acetone at room temperature. This operation was repeated for three times to completely remove unreacted PETA monomer and soluble homopolymer of PETA. The precipitated PiPP and PiPP-g-PETA were dried under vacuum at 80 °C for 24 h. The samples were pressed into a film (20 μm thick) and a sheet (1 mm thick) at 180 °C for further characterization.

Characterization

X-ray Photoelectron Spectroscopy Measurements. The chemical compositions of *i*PP and PiPP surfaces were characterized by the X-ray photoelectron spectroscopy (XPS; PHI-1600ESCA; PerkinElmer, Waltham, MA) with a monochromatic X-ray source (Al Ka). Contact angle was set at 45°. The analyzed area and vacuum pressure were 0.8 mm² and 10⁻⁷–10⁻⁸ Pa, respectively. The binding energy scale was referenced to the hydrocarbon C1s at 284.8 eV for all spectra.

Fourier Transform Infrared Spectrometry Measurements. The chemical structure of *i*PP and purified samples were characterized by the Fourier transform infrared (FTIR) spectrometer (Vertex 70; Bruker, Ettlingen, Germany) to calculate the GD. For all sample, 16 scans with a resolution of 4 cm⁻¹ were taken in the wavenumber range from 600 to 4000 cm⁻¹.

Rheological Measurements. The rotational rheometer (ARES-G2; TA Instruments, New Castle, DE) with a parallel plate (25 mm in diameter) was used for rheological measurements. The gap between plates was set at 0.8 mm. The small amplitude oscillatory shear experiments were performed in the frequency range of 0.03–500 rad s⁻¹ at 180 °C. The strain was fixed at 1%, which was in the linear viscoelastic regime for all samples.

RESULTS AND DISCUSSION

Air Plasma Modification of *i*PP Surface

Figure 1 shows the XPS spectrum of *i*PP and PiPP modified by air plasma. The corresponding quantitative information of carbon, oxygen, and nitrogen contents is summarized in Table I. From Figure 1 and Table I, it can be seen that the content of carbon on PiPP surface is less than on *i*PP and the O/C ratio shows a strong dependence on plasma treatment conditions. The oxygen on the surface of *i*PP is adsorbed oxygen (contamination). These results indicate that air plasma can break some of the C–C

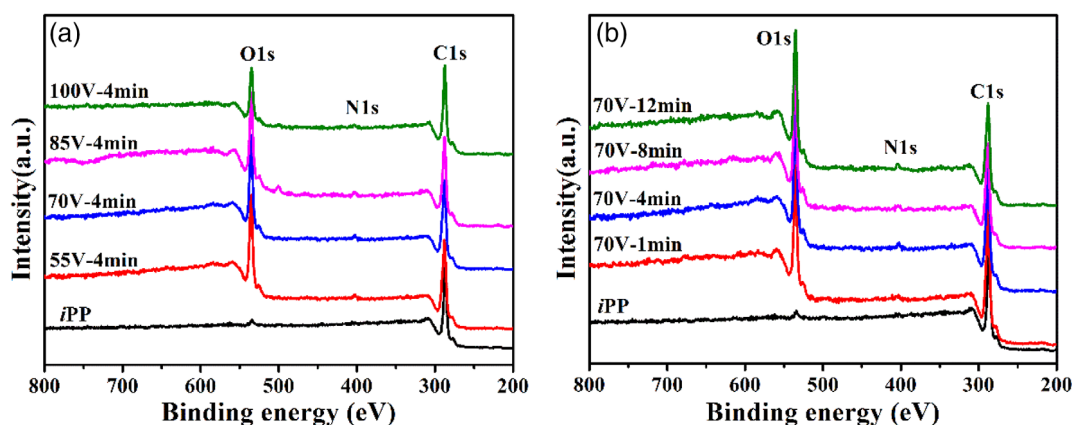


Figure 1. XPS spectrum of *i*PP and *Pi*PP modified by air plasma with (a) different discharge voltages for 4 min and (b) different times for 70 V. [Color figure can be viewed at wileyonlinelibrary.com]

Table I. XPS Elemental Analysis of *i*PP and *Pi*PP

Samples	Treatment condition		Atomic concentration ratio (%)			Component ratio (%)	
	Voltage (V)	Time (min)	C	O	N	O/C	N/C
<i>i</i> PP	0	0	97.3	2.7	0.0	2.8	0
<i>Pi</i> PP	55	4	74.6	24.3	1.2	32.6	1.6
<i>Pi</i> PP	70	4	70.1	28.2	1.7	40.2	2.5
<i>Pi</i> PP	85	4	71.1	27.1	1.7	38.1	2.4
<i>Pi</i> PP	100	4	77.8	20.5	1.7	26.3	2.2
<i>i</i> PP	0	0	97.3	2.7	0.0	2.8	0
<i>Pi</i> PP	70	1	72.8	25.3	1.9	34.8	2.6
<i>Pi</i> PP	70	4	70.1	28.2	1.7	40.2	2.5
<i>Pi</i> PP	70	8	66.7	31.2	2.1	46.8	3.1
<i>Pi</i> PP	70	12	66.9	31.1	2.0	46.5	3.0

bonds on the polymer surface, causing the polymer to recombine with the oxygen atoms. Under our experimental conditions, the maximum oxygen content is obtained with discharge voltage of 70 V and treatment time of 8 min, indicating that oxygen content on the surface reaches saturation. Although the treatment time is further increased, the oxygen content is almost unchanged.^{22,23} However, unlike the oxygen element, the small amount of nitrogen on the surface of *Pi*PP remains essentially unchanged. The reason is that the dissociation energy of N_2 molecules is higher than that of O_2 molecules, so that the density of nitrogen atoms in the plasma system is much lower than that of oxygen atoms. On the other hand, polymer surface treated also has a higher affinity for reactions with oxygen atoms than nitrogen atoms.^{24,25}

In order to investigate the chemical groups introduced on the *Pi*PP surface, the details of C1s peaks decomposition for *i*PP and *Pi*PP under different discharge voltage are summarized in Figure 2 and Table II. As shown in Figure 2(a), the C1s XPS spectrum of untreated *i*PP can be fitted with a single peak at around 284.8 eV, which can be attributed to C—H/C—C bonds. After the plasma treatment, some new peaks with higher binding energy appeared in the *Pi*PP spectrum. In Figure 2(b–e), the

other two new peaks appear at 286.5 and 288.2 eV, respectively. The peak located at 286.5 eV can be attributed to C—O bonds. The presence of this peak suggests that the *Pi*PP contains low-level oxidized carbon centers on the surface, such as alcohol, ether, and (hydro) peroxides. The peak located at 288.2 eV indicates the presence of chemical groups containing C=O bonds such as ketone and acetal.²⁶

As seen from Table II, data collected for *i*PP and *Pi*PP surfaces indicate the content of C—O and C=O bonds of the *Pi*PP surface increase first and then decreases with the increase of the plasma voltage. When the treatment voltage is 70 V, the content of the newly introduced C—O and C=O groups on the *Pi*PP surface reach the highest values of 24.4 and 12.0%, respectively. This indicates that for 4 min treatment time, the optimum treatment voltage is 70 V. When the treatment voltage is further increased, plasma energy becomes too high that its etching effect may happen on the surface of the *i*PP and consequently lower the content of the oxygen-containing functional group.

The high-energy particles of air plasma act on the surface of *i*PP to dehydrogenate and break the main chain of *i*PP, producing free radicals. When the *Pi*PP is exposed to air, these free radicals can be oxidized to introduce peroxides on the surface of the

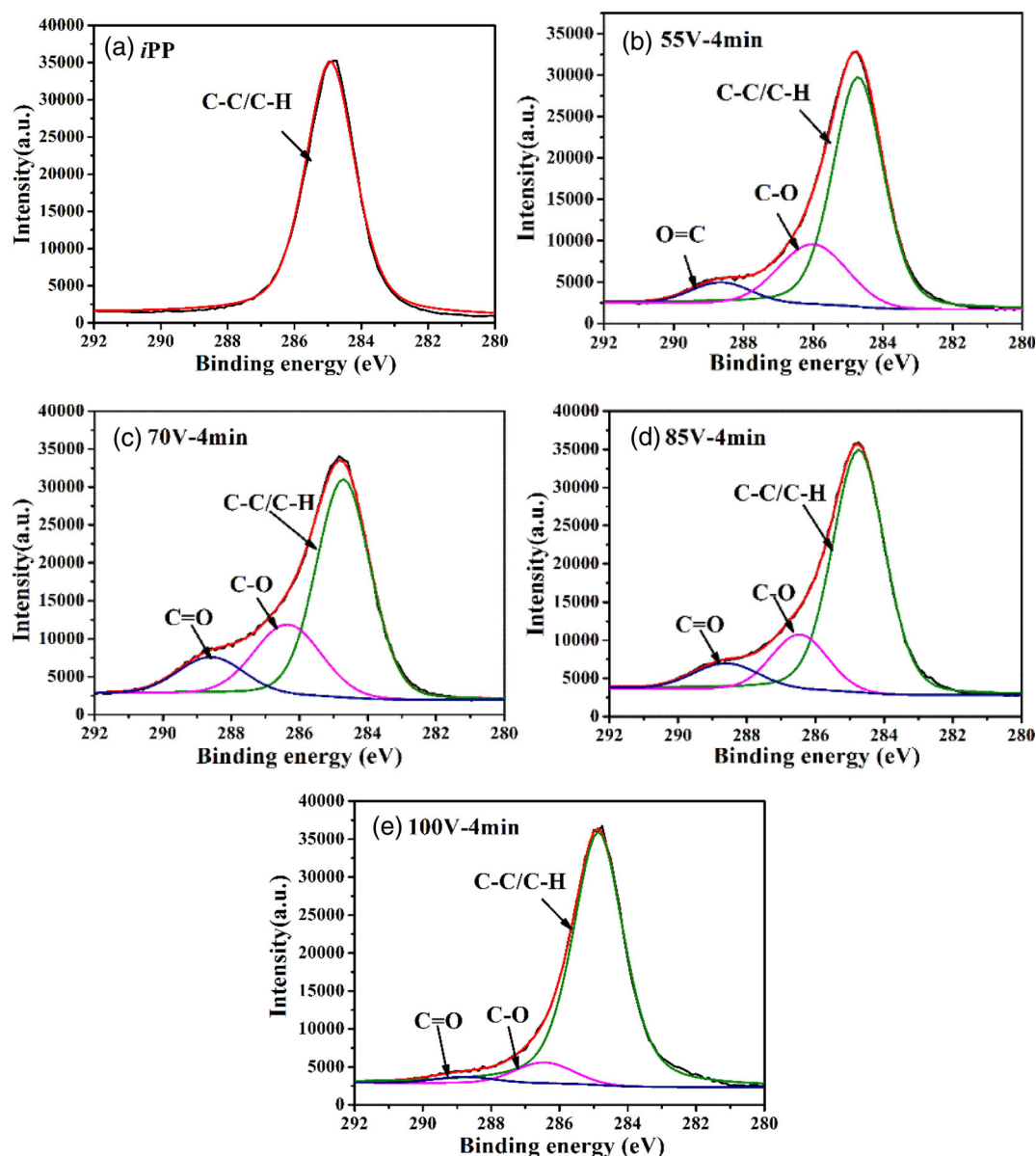


Figure 2. Carbon C1s XPS spectra of (a) *i*PP and PiPP by air plasma for (b) 55, (c) 70, (d) 85, and (e) 100 V. [Color figure can be viewed at wileyonlinelibrary.com]

materials.²⁷ These peroxides on the surface of the PiPP are effective active sites for inducing the grafting reaction. We attempted to detect peroxides on the PiPP by XPS, but even a qualitative analysis was not successful for the detection of peroxides. It was

reported that DPPH method can measure the concentration of peroxides.^{20,28} Therefore, we will use the DPPH method to determine the peroxides concentration generated on the polypropylene surface after plasma treatment.

Figure 3 shows the relation of the concentrations of peroxides with plasma discharge voltage and treatment time on the surface of PiPP. When the plasma treatment time is relatively short, for example, 1 or 2 min, the concentration of peroxides always increases with the increase of the discharge voltage. However, when the plasma treatment time is further increased, the peroxides concentration decreases after reaching the maximum value. As treatment time increases, free radicals are more likely to have a termination reaction. In particular, when the discharge voltage is also increased, etching effect occurs on the polypropylene surface, which also consumes a part of the free radicals. Therefore,

Table II. XPS Results of *i*PP and PiPP Treated with Different Discharge Voltages for 4 min

Adscription	Discharge voltage (V)				
	0	55	70	85	100
C—C/C—H(%)	100	71.1	63.6	74.9	91.0
C—O(%)	—	22.3	24.4	16.4	6.9
C=O(%)	—	6.6	12.0	8.7	2.1

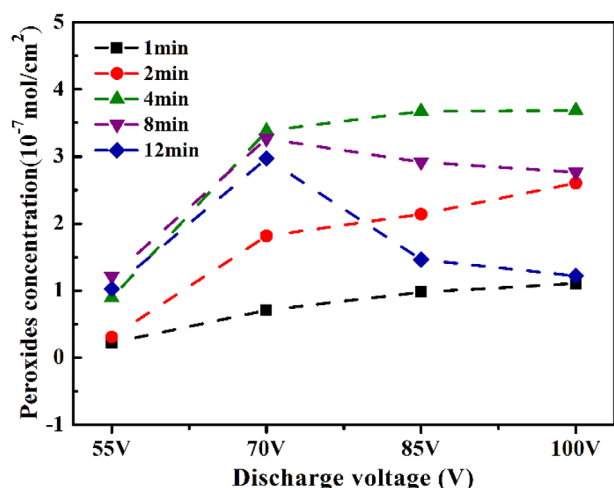


Figure 3. The relation of concentration of peroxides in PiPP surface with different discharge voltages and treatment times. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

the peroxides concentration also decreases. In summary, there is an optimum plasma treatment time of 4 min to obtain the high peroxides concentration of about $3.38\text{--}3.69 \times 10^{-7} \text{ mol cm}^{-2}$.

Graft Copolymerization

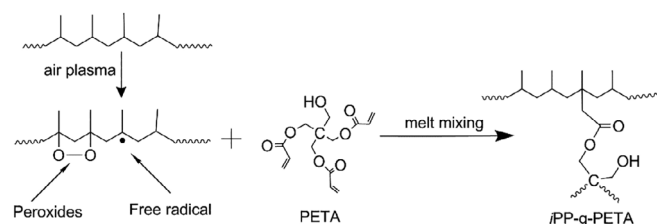
The peroxides on the PiPP inducing graft copolymerization of PETA onto PiPP are completed under melt mixing. The reaction process of PiPP with PETA during melt mixing is shown in Scheme 2.

For all samples, the plasma treatment time of iPP was 4 min with a fixed airflow rate of 100 mL min^{-1} , the effects of other experimental conditions including discharge voltage, mixing time, mixing temperature, and the concentration of PETA on the GD were investigated.

The GD was calculated by the FTIR of iPP and the purified samples (PiPP and PiPP-g-PETA). For all purified samples, such as Figure 4, a new absorption peak appeared around 1735 cm^{-1} , which is due to the stretching vibration of the carbonyl group of the ester in the PETA molecule.^{29,30} This indicates that the PETA molecule has been successfully grafted onto PiPP backbone. According to the FTIR spectrum, the absorbance ratio R_{CI} was calculated as follows:

$$R_{CI} = \frac{A_{1735}}{A_{841}} \quad (1)$$

where A_{1735} is the absorbance at 1735 cm^{-1} , characteristic of carbonyl groups of ester in the PETA molecule, and A_{841} is the



Scheme 2. The reaction process of PiPP with PETA during melt mixing.

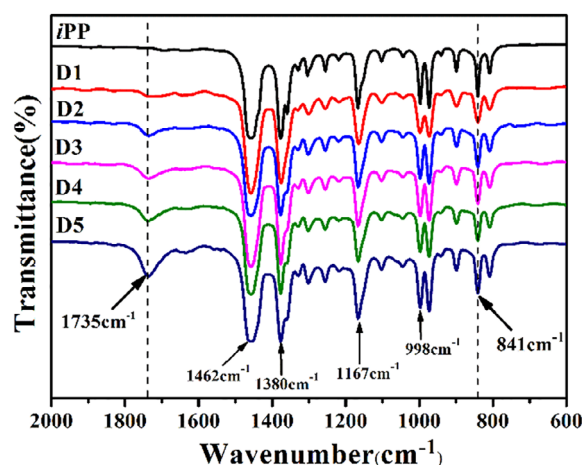


Figure 4. FTIR spectra of iPP and modified samples with different concentrations of PETA. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

absorbance at 841 cm^{-1} , characteristic of CH_3 groups in the iPP backbone. Since R_{CI} is a relative value, a calibration curve needs to be constructed to obtain the GD. A calibration curve is obtained by preparing the blends of iPP with assigned quantities of PETA, and that is presented in Figure 5. The GD of PETA onto PiPP, GD, at the different experimental conditions is listed in Table III and presented in Figure 6.

As shown in Figure 6(a), the GD of PETA increases rapidly with increasing discharge voltage from 55 to 100 V, and reaches a plateau after 85 V. The variation of discharge voltage reflects the variation of peroxides concentration on PiPP surface. Figure 3 shows that the change of peroxides content on PiPP surface becomes flat when discharge voltage is greater than 85 V, that is, the GD of PETA corresponds to the macromolecular peroxy radicals content in PiPP.

Next, the effects of mixing time and mixing temperature on the GD are studied in this work. In Figure 6(b), the GD of PETA increases with the increase of mixing time and reaches the

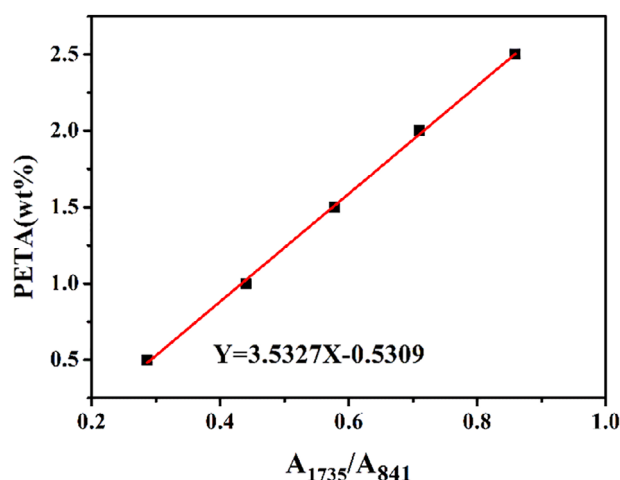


Figure 5. Calibration curve for the determination of the weight percentage of grafted PETA monomers. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

Table III. The GD of PETA onto PiPP under Melt Mixing at Different Experimental Conditions

Sample	PiPP (g)	PETA (g)	Discharge voltage (V)	Mixing time (min)	Mixing temperature (°C)	GD (wt %)
A1	100	5	55	10	180	0.15
A2	100	5	70	10	180	0.79
A3	100	5	85	10	180	1.22
A4	100	5	100	10	180	1.29
B1	100	5	70	10	180	0.79
B2	100	5	70	15	180	1.28
B3	100	5	70	20	180	1.24
C1	100	5	70	15	170	0.09
C2	100	5	70	15	180	1.28
C3	100	5	70	15	190	1.97
C4	100	5	70	15	200	1.76
D1	100	1	70	15	180	0.21
D2	100	2	70	15	180	0.49
D3	100	3	70	15	180	0.62
D4	100	4	70	15	180	1.04
D5	100	5	70	15	180	1.28

plateau after 15 min, indicating that the grafting reaction of PETA onto PiPP has finished with 15 min at 180 °C. Figure 6 (c) reveals the relationship between GD and mixing temperature. The GD of PETA initially increased and reached maximum at 190 °C followed by decrease with the further increase in the

reaction temperature. The initial increase in GD is mainly due to fact that both the complete decomposition constant and the reaction constant of the peroxides increase as the reaction temperature increases, where more free radicals were produced. On the other hand, the termination of free radicals increases with the

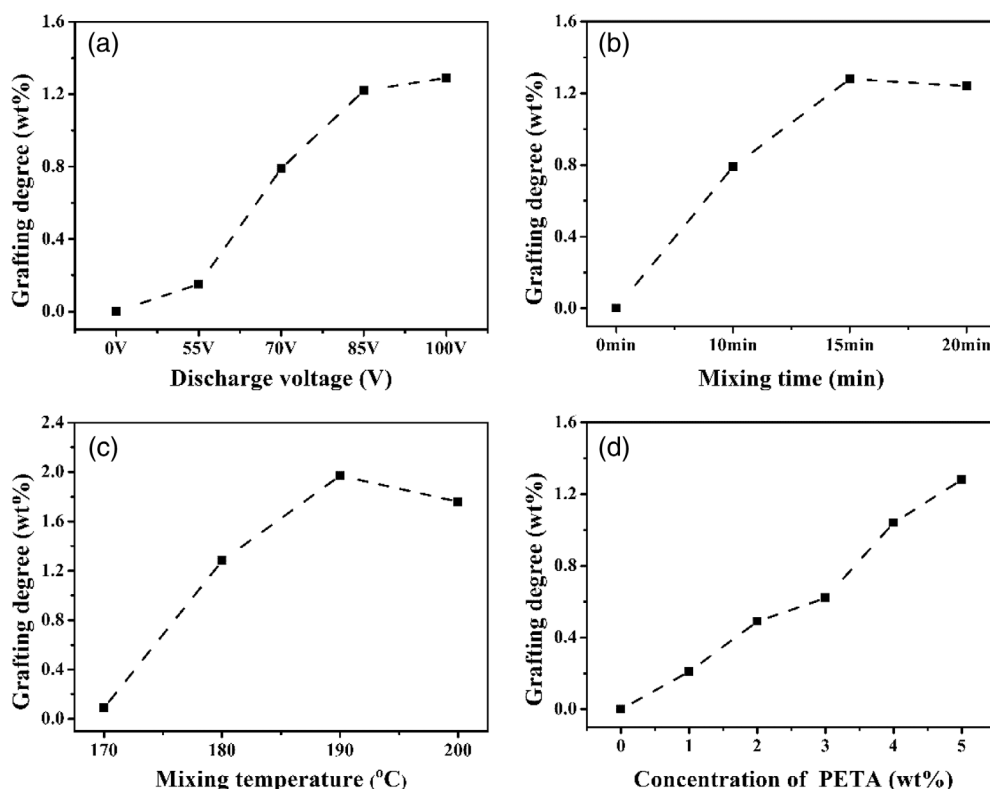


Figure 6. The GD of PETA onto PiPP under different experimental conditions (a) discharge voltage, (b) mixing time, (c) mixing temperature, and (d) the concentration of PETA.

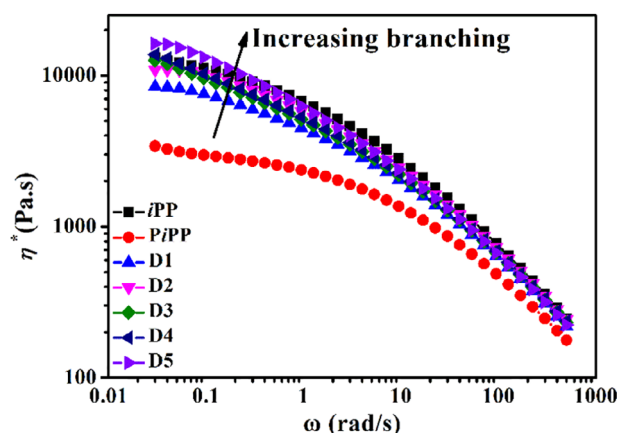


Figure 7. A plot of complex viscosity versus angle frequency for *i*PP, *Pi*PP, and modified *i*PP at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

increase of reaction temperature, and the probability of homopolymerization of PETA monomer also increases, resulting in a decrease in GD.³¹

From Figure 6(d), the GD of PETA increases with the concentration of PETA, suggesting that more PETA would favor capturing macromolecular peroxy radicals and grafting reaction.

These results show that the air plasma can induce grafting reaction of PETA onto polypropylene under melt mixing. The purpose of functional modification of polypropylene is realized, that is, polar groups are introduced into the polypropylene. Simultaneously, this modified polypropylene is also a graft of LCB polypropylene.

Linear Viscoelastic Properties

Linear viscoelastic properties are very sensitive to the structural change in the polymer. It has been demonstrated below from different rheological plots that the air plasma-induced grafting reaction of PETA onto *i*PP can result in new topological structure with longer relaxation time.

The presence of very low amounts of LCB can change the zero-shear viscosity (η_0) and the degree of shear thinning, as compared to the linear polymer with similar molecular weight.³² The complex viscosity (η^*) of *i*PP, *Pi*PP, and modified *i*PP at 180 °C is shown in Figure 7. The complex viscosity (η^*) of *Pi*PP is obviously decreased at low frequency compared to *i*PP, and Newtonian zone ranges became broader, showing that degradation

reaction happened and the molecular weight decreased. With the addition of PETA, the complex viscosity (η^*) of modified *i*PP (D1–D5) became higher than that of *Pi*PP at low frequency, indicating that the degradation was suppressed. With the increase of PETA, the complex viscosity (η^*) of modified *i*PP gradually increased at low frequency. The transition from the Newtonian plateau to the shear-thinning regime was shifted to lower frequency. The frequency dependence of complex viscosity (η^*) can be described by the cross equation,³³ which is written as

$$\eta^*(\omega) = \frac{\eta_0}{1 + (\lambda\omega)^n} \quad (2)$$

where η_0 is the Newtonian viscosity, λ is the cross time constant whose reciprocal accounts for the onset of shear-thinning region, and n is a shear-thinning index. These fitting results are summarized in Table IV. It can be seen that with increasing PETA content to 4%, η_0 and λ increase, while n decreases.¹⁷ The exception is that after adding 5% PETA, λ decreases and n increases, which could be associated with the possible reason that as the amount of PETA continues to increase, homopolymerization of the PETA monomer may occur. Although the GD increases, it is mainly reflected in the elongation of the graft chain and the decrease in graft density. This change in the graft structure results in a decrease in λ in the fitting result. Thereby, the effect of branches on viscosity is clear that there is no evident Newtonian plateau at low frequency and the shear thinning starts at lower frequency than *i*PP and *Pi*PP, which is consistent with the work done by Yu and coworkers.¹⁷ The complex viscosity (η^*) of modified *i*PP (D4 and D5) is higher than *i*PP at low frequency, but is lower than *i*PP at high frequency. These are typical characters of polymer with LCB.³⁴

Besides zero-shear viscosity (η_0), the storage modulus (G') and the loss angle (δ) are also sensitive to LCB.³⁰ In the terminal zone the storage modulus (G') and loss modulus (G'') of linear polymer follow the well-known frequency dependences, that is, the G' and G'' are proportional to ω^2 and ω when ω approaches zero for which only the longest relaxation times contribute to the viscoelastic behavior.³⁵ The G' of all samples plotted as a function of frequency is shown in Figure 8. The terminal slope of $\log G'$ versus $\log \omega$ is about 2 in the terminal zone for linear polymer. The terminal behavior of *i*PP and *Pi*PP is exhibited to indicate that they are linear polymers. The commercial *i*PP is not a typical linear polymer, and so the terminal slope of $\log G'$ versus $\log \omega$ is not about 2. The terminal slope of G' is 1.19 here, and the

Table IV. Rheological Parameters of *i*PP, *Pi*PP, and Modified *i*PP

Sample	GD (wt %)	$\eta_{0.03 \text{ rad s}^{-1}}^*$ (Pa s ⁻¹)	$G'_{0.03 \text{ rad s}^{-1}}$ (Pa)	Terminal slope of G'	η^0 (Pa s ⁻¹)	λ	n	Bn
<i>i</i> PP	—	13,071	57.77	1.19	14,671	1.22	0.57	—
<i>Pi</i> PP	—	3406	—	1.29	3197	0.17	0.59	—
D1	0.21	8418	58.72	1.19	10,521	1.70	0.53	—
D2	0.49	10,891	79.73	1.12	14,085	1.86	0.50	—
D3	0.62	12,591	128.63	0.92	18,419	7.70	0.45	0.011
D4	1.04	13,769	154.13	0.87	21,543	11.44	0.44	0.018
D5	1.28	16,158	160.88	0.87	25,325	8.86	0.50	0.026

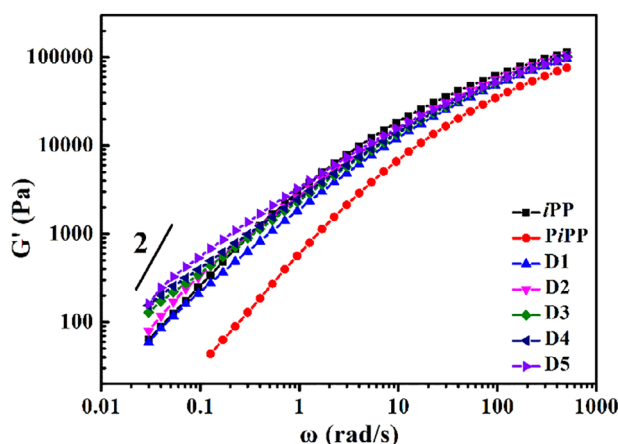


Figure 8. Storage modulus versus angle frequency for *i*PP, PiPP, and modified *i*PP at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

terminal slope of PiPP is 1.29. The other five samples (D1–D5) modified with PETA are deviated from the terminal behavior as compared to PiPP.

With increase of PETA, G' of modified *i*PP increases at low frequency, the terminal slope of G' decreases from 1.19 of *i*PP to 0.87 of D5. The nonterminal behavior of D1–D5 indicates that there is a longer relaxation process, which can be ascribed to the LCB formed from plasma-induced melt grafting reaction of PETA onto PiPP.

Figure 9 exhibits the relation of $\tan \delta$ ($\tan \delta = G''/G'$) with angle frequency (ω) for *i*PP, PiPP, and modified *i*PP. The $\tan \delta$ increases with a decrease in ω for linear polymers, that is, *i*PP and PiPP. With the addition of PETA, $\tan \delta$ of modified *i*PP decreases quickly at low frequency and shows a plateau. If more PETA was added, the $\tan \delta$ decreased continually and the plateau became broader, indicating the LCB on the *i*PP skeleton increase the terminal relaxation time. This observation is in accordance with the result of Wood-Adams and Dealy.³⁶

A significant influence of LCB on linear rheological behavior can be seen in a plot of the phase angle (δ) versus the absolute value

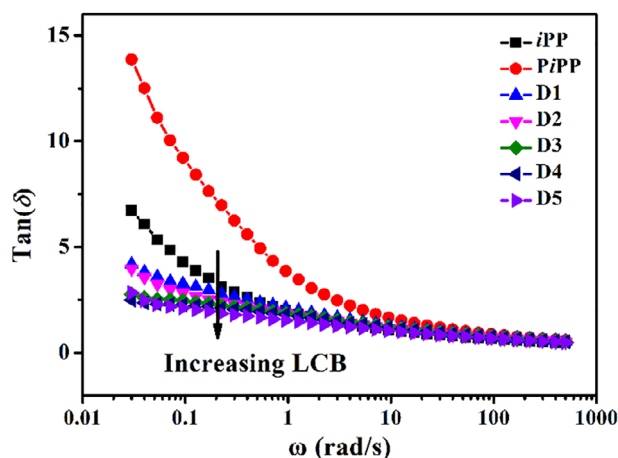


Figure 9. $\tan \delta$ versus angle frequency for the linear *i*PP, PiPP, and modified *i*PP samples at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

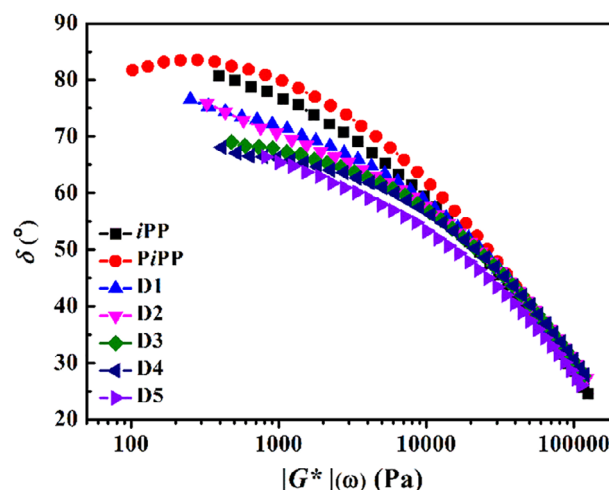


Figure 10. Variation of phase angle, δ , as a function of complex modulus, $|G^*|$, for *i*PP, PiPP, and modified *i*PP at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

of complex modulus ($|G^*|$) often called the van Gurp–Palmen plot.³⁷ In Figure 10, the values of δ for *i*PP and PiPP are nearly 90° at the low values of $|G^*|$. The δ – $|G^*|$ curves of modified *i*PP appear pronounced deviations as compared to linear polymers, which becomes more prominent with increasing amount of PETA. Polymers are almost completely viscous when the δ value is close to 90° and almost completely elastic when the δ value is close to 0°. Therefore, the elasticity of modified *i*PP is higher than that of *i*PP and PiPP, and the elasticity of modified *i*PP increases with LCB increasing.

The nonterminal behavior of *i*PP and PiPP samples can be illustrated in the Han plot ($\log G'$ and $\log G''$, Figure 11). The Han plot has been used to investigate order–disorder transitions in block copolymers,^{32,38} the effect of polydispersity and long branched chain in polyethylene,³⁹ and miscibility of polymer blends.^{40,41} Han plot is independent of the melt temperature and weight-average molecular weight for monodispersed polymers. As seen from Figure 11, the Han plots of the modified *i*PP

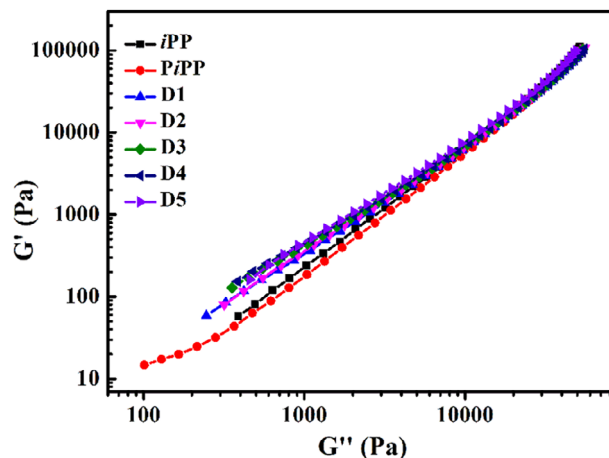


Figure 11. Han plot of the *i*PP, PiPP, and modified *i*PP at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

samples deviate up obviously from the plot of linear polymer, showing that a long relaxation time occurred in these samples. This long relaxation process can also be seen in the Cole–Cole plot.

The Cole–Cole plot is used to determine the phase separation of the blends⁴² and to characterize the rheological properties of long chain branched polymers.⁴³ The Cole–Cole curve of linear polymer is nearly semicircular, of which the radius increases with molecular weight.

In Figure 12, the radius of *i*PP Cole–Cole curve is much larger than that of *Pi*PP, because degradation of *Pi*PP leads to the drop of molecular mass. The Cole–Cole plot of the modified *i*PP is not semicircular. The Cole–Cole plots of D1 and D2 samples are lower than *i*PP, but higher than that of *Pi*PP, showing an increase in molecular weight after grafting. The Cole–Cole plots of D3–D5 samples are higher than *i*PP and show more evident upturning at high viscosity. It is clear that these linear viscoelastic plots exist a longer relaxation process in the modified *i*PP that results from air plasma induced radical reaction in the presence of PETA.

Quantitative Analysis of LCB

Obviously, above results only prove the existence of long-chain branch in the qualitative manner. Quantifying the LCB level of polymers is not an easy task, and even more difficult for our samples due to the complexity of the plasma-induced melt grafting reaction. Compared to NMR and gel permeation chromatography, rheology is a more appropriate technique for verifying the existence of LCB, especially for low level LCB far below the detection limit of the above methods.^{35,36,44} Tsenoglou and Gotsis⁴⁵ have studied the dependence of the zero-shear viscosity on the degree of branching for PP. The weight fraction of branched chains can be expressed as⁴¹:

$$Bn \approx \frac{\ln(\eta_{BL}/\eta_L)}{\alpha(M_L/M_C - 1) - 3 \ln(M_L/M_C)} \quad (3)$$

where η_{BL} is the zero-shear viscosity of the blend containing branched and linear polymers, η_L and M_L are the zero-shear viscosity and the weight-average molecular weight of the linear precursor, and the coefficient α is independent of the branching point functionality

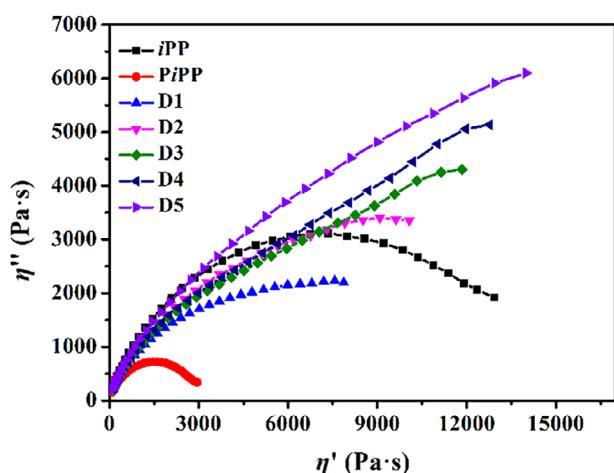


Figure 12. Cole–Cole plot of the *i*PP, *Pi*PP, and modified *i*PP at 180 °C. [Color figure can be viewed at wileyonlinelibrary.com]

($\alpha=0.42$). M_C is the molecular weight between two successive entanglements, which is around 11.2 kg mol^{-1} for PP. M_L was calculated from the relationship between zero-shear viscosity and the molecular weight of the linear PP, given as follows¹⁶:

$$\log \eta_0 = -15.4 + 3.5 \log M_L \quad (4)$$

where M_L is equal to 388 kg mol^{-1} , which is calculated from eq. (4). It should be noted here that eq. (3) can be used only when η_{BL} is larger than η_L . Therefore, the weight fraction of branched chains of D1 and D2 cannot be calculated because it has lower zero-shear viscosity than that of the linear precursor. The values of Bn for D3–D5 are calculated to list in Table IV. The result shows that the degree of branch (Bn) depends on PETA concentration.

From above discussion, it is clear that a longer relaxation process appeared in the modified *i*PP with LCB. This LCB structure is obtained by air plasma-induced melt grafting of PETA onto polypropylene.

CONCLUSIONS

In this work, the surface modification of polypropylene by DBD of air plasma was investigated. The results of XPS indicate that a large number of oxygen-containing functional groups such as C–O and C=O on the surface of the polypropylene after plasma treatment. The optimum conditions for plasma treatment are a discharge voltage of 70 V and a treatment time of 4 min. Under these conditions, the oxygen content of the polypropylene surface is as high as 28.2%; the O/C ratio is 40.2%. The corresponding peroxides concentration is $3.38 \times 10^{-7} \text{ mol cm}^{-2}$, which can effectively induce the melt grafting of polypropylene. Unexpectedly, the content of oxygen-containing functional group on polypropylene surface by air plasma treatment is higher than that obtained with argon and ethylene plasma.^{26,46}

The effects of plasma treatment conditions and grafting conditions on the melt grafting of PETA onto polypropylene were discussed. The results show that under the conditions of 180 °C and 60 rpm, the complete reaction of PETA and polypropylene takes about 15 min. Moreover, the highest GD can reach 1.79%, comparable with the chemical grafting method.^{14,17} Finally, the existence of the LCB structure in the graft modified polypropylene was confirmed and quantitatively determined by shear rheological measurements. The amount of LCB produced by the plasma method increases with increasing PETA concentrations, which is consistent with the chemical method like addition of peroxides.¹⁷

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (No. 51473113) and the Natural Science Foundation of Tianjin (No. 12JCYBJC11900).

REFERENCES

- Cho, J. D.; Kim, S. G.; Hong, J. W. *J. Appl. Polym. Sci.* **2010**, *99*, 1446.
- Amintowlieh, Y.; Tzoganakis, C.; Penlidis, A. *J. Appl. Polym. Sci.* **2014**, *131*, 41021.

3. Chen, H.; Ma, W.; Xia, Y.; Gu, Y.; Cao, Z.; Liu, C.; Yang, H.; Tao, S.; Geng, H.; Tao, G.; Matsuyama, H. *Appl. Surf. Sci.* **2017**, *419*, 259.
4. Akbar, D. *Appl. Surf. Sci.* **2016**, *362*, 63.
5. Moses, K. J.; Kim, S.; Bilal, M.; Cohen, Y. *J. Appl. Polym. Sci.* **2018**, *135*, 46843.
6. Bahrami, N.; Nouri Khorasani, S.; Mahdavi, H.; Ghiaci, M.; Mokhtari, R. *J. Appl. Polym. Sci.* **2019**, *136*, 47567.
7. Altay, L.; Bozaci, E.; Atagur, M.; Sever, K.; Tantug, G. S.; Sarikanat, M.; Seki, Y. *J. Appl. Polym. Sci.* **2019**, *136*, 47131.
8. Chiang, W. Y.; Ku, Y. A. *Polym. Degrad. Stab.* **2002**, *76*, 281.
9. Saxena, S.; Ray, A. R.; Gupta, B. *J. Appl. Polym. Sci.* **2010**, *116*, 2884.
10. Ma, G. Q.; Zhai, J. J.; Liu, B.; Huang, D. H.; Sheng, J. *Chinese J. Polym. Sci.* **2012**, *30*, 423.
11. Khelifa, F.; Ershov, S.; Habibi, Y.; Snyders, R.; Dubois, P. *Chem. Rev.* **2016**, *116*, 3975.
12. Gul Dincmen, M.; Hauser, P. J.; Gursoy, N. C. *Plasma Chem. Plasma P.* **2016**, *36*, 1377.
13. Luo, Z.; Chen, H.; Xu, J.; Guo, M.; Lian, Z.; Wei, W.; Zhang, B. *J. Appl. Surf. Sci.* **2018**, *135*, 46171.
14. Wan, D.; Xing, H.; Zhang, Z.; Wang, Y.; Wang, L.; Wang, Y.; Jiang, Z.; Tang, T. *J. Appl. Polym. Sci.* **2013**, *127*, 1394.
15. Calderón, B. A.; Soule, J.; Sobkowicz, M. J. *J. Appl. Polym. Sci.* **2019**, *136*, 47553.
16. Meemusaw, M.; Maia, J.; Jamieson, A.; Magaraphan, R. *Mater. Chem. Phys.* **2017**, *199*, 34.
17. Tian, J.; Yu, W.; Zhou, C. *Polymer.* **2006**, *47*, 7962.
18. Li, S.; Xiao, M.; Wei, D.; Xiao, H.; Hu, F.; Zheng, A. *Polymer.* **2009**, *50*, 6121.
19. Diop, M. F.; Torkelson, J. M. *Polymer.* **2015**, *60*, 77.
20. Suzuki, M.; Kishida, A.; Iwata, H.; Ikada, Y. *Macromolecules.* **1986**, *19*, 1804.
21. Hidzir, N. M.; Hill, D. J. T.; Taran, E.; Martin, D.; Grøndahl, L. *Polymer.* **2013**, *54*, 6536.
22. Greene, G.; Yao, G.; Tannenbaum, R. *Langmuir.* **2003**, *19*, 5869.
23. Cui, N. Y.; Brown, N. M. D. *Appl. Surf. Sci.* **2002**, *31*, 189.
24. Huertas, S. P.; Terpiłowski, K.; Tomczyńska-Mleko, M.; Mleko, S.; Szajnecki, Ł. *Colloid Surf. A.* **2018**, *558*, 322.
25. Šourková, H.; Primc, G.; Špatenka, P. *Materials.* **2018**, *11*, 885.
26. Ma, G. Q.; Liu, B.; Li, C.; Huang, D.; Sheng, J. *Appl. Surf. Sci.* **2012**, *258*, 2424.
27. Meyer-Plath, A. *Vak. Forsch. Prax.* **2010**, *17*, 40.
28. Dorey, S.; Gaston, F.; Marque, S. R. A.; Bortolotti, B.; Dupuy, N. *Appl. Surf. Sci.* **2018**, *427*, 966.
29. Ma, G. Q.; Sun, G. K.; Ma, Z.; Li, J. Q.; Sheng, J. *Chinese J. Polym. Sci.* **2018**, *36*, 979.
30. Wang, K.; Wang, S.; Wu, F.; Pang, Y.; Liu, W.; Zhai, W.; Zheng, W. *J. Mater. Sci.* **2016**, *51*, 2705.
31. Moad, G. *Prog. Polym. Sci.* **1999**, *24*, 81.
32. Malmberg, A.; Gabriel, C.; Steffl, T.; Münstedt, H.; Löfgren, B. *Macromolecules.* **2002**, *35*, 1038.
33. Borsig, E.; Van Duin, M.; Gotsis, A. D.; Picchioni, F. *Eur. Polym. J.* **2008**, *44*, 200.
34. Wood-Adams, P. M.; Dealy, J. M.; deGroot, A. W.; Redwine, O. D. *Macromolecules.* **2000**, *33*, 7489.
35. Ruymbeke, E. V.; Steohenne, V.; Daoust, D. *J. Rheol.* **2005**, *49*, 1503.
36. Wood-Adams, P. M.; Dealy, J. M. *Macromolecules.* **2000**, *33*, 7481.
37. Van Gurp, M.; Palmen, J. *Rheol. Bull.* **1998**, *67*, 5.
38. Zheng, Q.; Yang, B. B.; Wu, G.; Li, L. W. *Chem. J. Chinese U.* **1999**, *20*, 1483.
39. Choi, S.; Chang, D. H. *Macromolecules.* **2004**, *37*, 215.
40. Antony, P.; Puskas, J. E.; Kontopoulou, M. *Polym. Eng. Sci.* **2003**, *43*, 243.
41. Vega, J. F.; Santamaría, A.; Muñoz-Escalona, A.; Lafuente, P. *Macromolecules.* **1998**, *31*, 3639.
42. Zheng, Q.; Du, M.; Yang, B.; Wu, G. *Polymer.* **2001**, *42*, 5743.
43. Vittorias, I.; Wilhelm, M. *Macromol. Mater. Eng.* **2007**, *292*, 935.
44. Shroff, R. N.; Mavridis, H. *Macromolecules.* **2001**, *34*, 7362.
45. Tsenoglou, C. J.; Gotsis, A. D. *Macromolecules.* **2001**, *34*, 4685.
46. Ma, G. Q.; Liu, Y. P.; Wei, S. X.; Sheng, J. *Chinese J. Polym. Sci.* **2015**, *33*, 669.